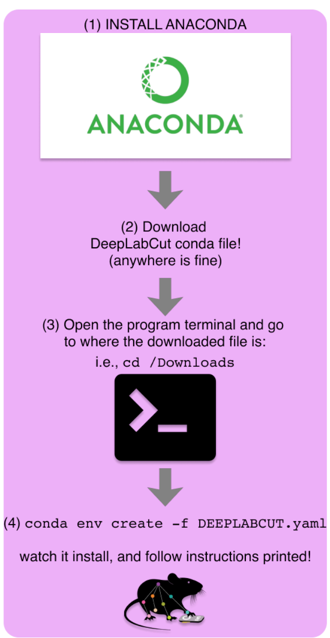
DeepLabCut and Anipose 3D Pen Tracking

In this activity, we will track the tip of a pen in 3D using a 3-camera motion tracking setup. DeepLabCut will be used to track the pen in each 2D camera view. Anipose will be used to triangulate 3D coordinates based on the three perspectives, and information about the relative location of each camera gained from calibration.

Deeplabcut

Downloading the Software

1. Go to https://deeplabcut.github.io/DeepLabCut/docs/installation.html and follow steps
2. Follow the Conda Environment method of installing that software
    a. First, verify that you have Anaconda or Miniconda installed on your device
    b. Download the Conda file provided by the DeepLabCut documentation where it says click HERE
    c. Navigate to the folder where you downloaded the file in the Terminal `cd C:\Users\YourUserName\Downloads`
    d. Run `conda env create -f DEEPLABCUT.yaml` in the terminal
3. 

Filming Your Own Videos

1. Film a 15-30 second clip of yourself moving the provided pen using one of the GoPros with the following settings: 1080p, 30 fps, linear. Cover a wide range of orientations and positions, but keep the tip of the pen visible to the camera. Stay 1-2 feet away from the camera. Avoid moving the pen too quickly, as the resulting motion blur will make tracking the pen harder.
2. Transfer the pen video from the camera's SD card to your computer. The video files will be in the DCIM folder. You only need the .MP4 file. Place this video in its own folder. The extension will likely be .MP4. Change it to .mp4 (filetypes in DeepLabCut and Anipose are case sensitive).

Creating the Project

1. Start the conda environment by running the following line in the terminal:

```
conda activate DEEPLABCUT
```

2. Before creating the project, you will want to downsample your video to decrease processing time. First start a python interpreter by typing 'python', 'ipython', or 'python3' into the terminal. Then, run the following lines, replacing "video path" with the path to the video you just filmed. To get this path on MacOS, right click on the video, hold option to bring up new options in the right click menu, then copy as pathname.

```
import deeplabcut
deeplabcut.DownSampleVideo("video path", width=-1, height=250)
```

3. Quit python by typing 'quit' or 'quit()', then run the following line to pull up the DeepLabCut GUI:

```
pythonw -m deeplabcut
```

4. Create a new project with a descriptive name (in this case, "Pen" or "Pen Tip" work fine). Click 'Load Videos' and select your downsampled pen video. Enter your name or pseudonym into the 'Name of the experimenter' field.
5. Under optional attributes, check the first box to change where the DLC project file is created if desired (by default, it will be created inside your user profile directory).
6. Click 'Ok' to create the project folder, then 'Edit config file' to pull up the project's configuration file. Make sure the default app set to open the config file is a text editor so you can edit the parameters.

Configuration

1. Inside the config file, you will want to change the 'bodyparts' and 'skeleton' parameter. 'bodyparts' defines the points to be tracked, while 'skeleton' defines how they are

connected. You are only tracking the tip of the pen, so there should only be one bodypart and no skeleton.

```
bodyparts:
- tip
skeleton:
```

2. 'numframes2pick' determines how many frames DeepLabCut will extract from your videos for labeling. Raise this to 30.

Labeling

1. Move to the 'Extract Frames' tab, then click 'Ok'. This will create a folder containing frames automatically extracted from your pen video at the location [Project File Path]/labeled-data. Automatic extraction may take several minutes.
2. Move to the 'Label Data' tab, then click 'Label Frames' to pull up the labeling GUI. Load frames, then select the folder containing the extracted frames for your video. The first extracted frame should now be displayed in the GUI.
3. Navigate between the frames using the '<<Previous' and 'Next>>' buttons, and label the tip of the red pen in each frame by <u>right clicking</u>. Use the Zoom tool to apply your labels more precisely. Adjust the marker size in the top right if desired (we find a marker size of 2 works well). Once all frames are labeled, save then quit the labeling GUI.

Training

4. Move to the 'Create Training Dataset' tab, hit 'Ok', then move to the 'Train' tab. Under 'Want to edit pose_cfg.yaml file?', click Yes. This should pull up another config file, pose_cfg.yaml. If not, you can find this file manually at the location [Project File Path]/dlc-models/iteration-#/{Project Name}-trainset95shuffle1/train.
5. Here you want to change the parameters as follows:

```
display_iters: 100
save_iters: 500
```

6. Now, click 'Update the parameters'. display_iters determines how often DLC prints out training progress to the terminal. As it is now, DLC will print progress every 100 iterations. save_iters determines after how many iterations DLC will save a model, with DLC saving every 500th iteration.
7. Click 'Ok' to begin training, and let the model train.
8. **Start on the Anipose section, and proceed up through the Filming section.**
9. After finishing up through Filming in the Anipose section, stop training with Cmd/C or Ctrl/C (depending on OS). Doing so will also quit DeepLabCut, so re-open the GUI:

```
pythonw -m deeplabcut
```

10. Load your project by selecting the project's config file.

Results

1. On the next tab, you will evaluate your model's performance. Click 'RUN: Evaluate Network', and view the results in the terminal. The given error represents the average pixel distance between your labels and the model's labels.
2. Now it is time to test the model out on your video, as well as videos that were not part of the training set. Download a the example pen tracking video here, and head to the 'Analyze videos' tab. Select your downsampled pen video and the one you downloaded. Then, disable and re-enable plotting trajectories (for some reason plots will appear empty unless you do this). Now, run the analysis. This will create data files in the same directory as their corresponding video.
3. In the 'Create videos' tab, specify the correct videotype (.mp4 in our case), select the videos you analyzed in the previous step, then click 'RUN' to create your labeled videos. These should appear in the same directory as the original videos.

Anipose

We will be tracking the pen in 3D space using a 3-camera setup. Anipose uses a trained DeepLabCut pen model to track the pen across the three camera views in 2D, then estimates 3D points by triangulating between the three views given the relative positions of the cameras, found through a calibration step. We will use a pre-trained DeepLabCut pen model for this section.

Setup

1. You will be running Anipose on a Google Colab notebook to decrease processing time. Download the folder called 'Anipose-Pen-Tip' here. The folder will be your Anipose project. Also download the DeepLabCut project called 'Pen Tip-Sotolab-2022-07-20'. Place these somewhere in your Google Drive.

Filming

1. Camera Setup
    a. For 3D tracking, all of the cameras must be positioned facing the desired object. We have placed Cameras A, B, and C in alphabetical order from left to right.
    b. A reference photo is provided below:

Reference Photo for Camera Setup

2. Pen Filming
    a. While all three cameras are recording, move the pen at a slow to moderate speed in the view of the camera. Avoid moving the pen too quickly, as the resulting motion blur will make tracking the pen harder.
3. Checkerboard Filming
    a. While the cameras are recording, move the checkerboard around the recording area, making sure to stay within view of all three cameras. Try to cover a large portion of the desired recording area.
    b. Tilt the checkerboard slightly forwards and backwards at various points in the view frame, not exceeding a 45 degree angle.
    c. An example calibration video can be found here.
4. Transfer the calibration and pen videos to the Anipose-Pen-Tip folder in your Drive. Place and name them as follows:

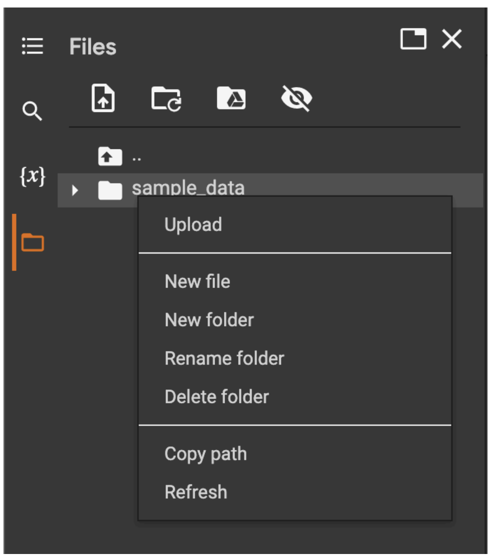
```
Anipose-Pen-Tip
- Cam A
  - pen-CamA.mp4
- Cam B
  - pen-CamB.mp4
- Cam C
  - pen-CamC.mp4
- Calibration
```

```
- calibA.mp4
- calibB.mp4
- calibC.mp4
```

5. At this point, go back to step 9 in the DeepLabCut Training section.

3D Tracking

1. Open the Colab notebook here. In the top bar, navigate to Runtime > Change runtime type, then change your hardware accelerator to the GPU. Connect to the runtime (upper right corner). This notebook will triangulate between your three camera views, then generate a .gif file plotting the pen's 3D coordinates.
2. Run the cells up through the one that mounts your Google Drive to Colab. Restart runtime if you are prompted to do so.
3. Navigate to the configuration cell (should be the 4th cell). You will need to edit the ANIPOSEPROJECT and DEEPLABCUTPROJECT variables to be the locations of your Anipose and DeepLabCut project folders (the ones you downloaded during the setup). To easily find the path of a folder in your Drive, find them in the file navigator on the left hand side, and right click on the folder to bring up the "Copy path" option.



Make sure the other settings match the ones below.

```
VIDEOTYPE = 'mp4'
FRAMERATE = 30
CAMREGEX = 'Cam[ABC]'

DOWNSAMPLE = True
```

```
DOWNSAMPLEWIDTH = 444

CALIBRATE = True
CALIBRATIONCAMA = 'calibA.mp4'
CALIBRATIONCAMB = 'calibB.mp4'
CALIBRATIONCAMC = 'calibC.mp4'

BOARDTYPE = 'checkerboard'
BOARDDIMX = 8
BOARDDIMY = 6
SQUARELENGTH = 1

SCHEME = []
```

4. Now, with the configuration cell selected, click Runtime > Run after. This will run all the cells after the configuration cell.

Results

1. Check the Anipose project folder, and you will see a new folder called 3D Data. In here, you will find .csv files containing 3D coordinates, and a .gif file plotting those coordinates.